Kwang Myung Yu

AI Engineer · Director, R&D Group, GenON

01 PROFILE

Started in power plant automation and model predictive control, then developed anomaly detection, forecasting, and computer vision models for energy and manufacturing. Led RAG, agent, and Text-to-SQL services for enterprise clients, and now leads GenON's Physical AI humanoid product, GenP, from product definition to VLA training pipelines.

02 SKILLS

Programming — Python for ML and services, C/C++ for control and embedded code, plus SQL and TypeScript.

AI/ML — LLM agents and RAG with LangChain/LangGraph, LiteLLM, OpenRouter, FastMCP, and Braintrust; anomaly detection, time-series forecasting, and computer vision with PyTorch and scikit-learn.

Robotics — Humanoid development on Unitree G1 and AgiBot X2 SDKs, VLA training, simulation in Isaac Lab and MuJoCo, and SLAM-based navigation.

Serving & Infrastructure — Model and LLM serving with FastAPI, vLLM, and Ollama on Docker and AWS, with MLflow, PostgreSQL, and vector databases (pgVector, Qdrant).

Domains — Energy and utilities (thermal power, renewables, ESS), petrochemical and steel processes, manufacturing, healthcare, and industrial automation (PLC, DCS).

03 EXPERIENCE

GenON, R&D Group

Jul 2026 - Present

Director / AI Product Engineer

Lead the Physical AI humanoid product as Product Owner (PO), defining the product, technology tree, and commercialization roadmap, and manage client projects as Project Manager (PM). Design the integrated architecture for voice conversation, scenario execution, VLA, VLN, navigation, and motion, and build the robot control panel. Propose and run new Physical AI and agent-system projects and hire talent.

GenP Physical AI Humanoid Product

2026 - Present

- Own the product as PO, writing the product definition and commercialization roadmap around a common core + domain profiles structure.
- Architect a single workflow that integrates voice conversation, VLA manipulation, SLAM navigation, and whole-body control, and design and build a control panel for robot registration, monitoring, logs, alarms, task execution, and emergency stop.
- Lead the data and model pipeline, from Teleop data collection to VLA training and evaluation and a simulation environment for validation.

Unitree G1 · Unitree SDK · LeRobot (VLA) · Isaac Lab · MuJoCo · FAST-LIO2 · FastAPI · React · Next.js

Docent Humanoid Robot for a Brand Experience Center

Planned

2026 - 2027

Global F&B company

- Manage the project as PM, defining scope, WBS, and acceptance criteria and running kickoff and interim reviews.
- Design the architecture for voice conversation (STT/TTS), tour scenarios, SLAM-based autonomous navigation, and a LiDAR/depth collision-avoidance safety system.
- Build a control dashboard for robot status and alarms, scenario execution, language settings, and E-Stop.

AgiBot X2 Ultra · AgiBot SDK · SLAM · FastAPI · React · Next.js · External STT/TTS APIs



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WEB sguys99.github.io

EDUCATION

M.S. in Electrical Engineering
KAIST · 2006 - 2008

B.S., School of Electrical Engineering
Pusan National University · 1999 - 2006

AWARDS

Best People Awards
PwC Consulting · 2025

President's Award (Excellence Prize), Big Data & AI Competition
Korea East-West Power (EWP) · 2020

Merit Award (R&D)
KEPCO · 2018

Honors graduate; academic excellence scholarship
Pusan National University · 2000, 2003 - 2006

PwC Consulting, Digital & AI Mar 2024 – Aug 2025

Director / Senior ML Engineer

Developed generative AI services (RAG and agents), ML-based equipment diagnostics, price and demand forecasting, and control optimization. Proposed new projects, ran AI training for firm-wide and division staff, and hired talent.

> **SCM Data Analysis and Work Support Service** 2025
>
> Samsung Electronics
>
> – Led the project end to end and designed the service architecture and agent workflows built on Text-to-SQL.
> – Deployed the service and managed its operation.
>
> Qwen 8B (fine-tuned) · Llama 3.1 70B · LangChain · LangGraph · Hugging Face · vLLM · Milvus · LiteLLM · Braintrust
>
> **Online Consultation and Product Recommendation Assistant** 2024 – 2025
>
> Samsung Electronics
>
> – Led the project end to end, designed the service architecture and agent/RAG workflows, and deployed and operated the service on Azure.
> – Established an evaluation framework for the RAG service and built a program to automate evaluation.
>
> Microsoft Azure · Azure AI Search · Python · LangChain · RAGAS
>
> **Raw-Material Price Forecasting and Market Analysis Dashboard** 2024
>
> LOTTE Fine Chemical
>
> – Framed price forecasting as supervised regression over about 150 candidate attributes and compared Linear Regression, Random Forest, XGBoost, and LSTM against a rule-based benchmark.
> – Cut forecast error (MAPE) from 5.9% with the rule-based method to 1.3%, with multi-horizon forecasts 1–4 weeks and 1–3 months ahead.
> – Built a four-page dashboard (Upload, EDA, Training, Dashboard) with SHAP-based driver analysis, letting business users upload data and retrain models themselves.
>
> XGBoost · scikit-learn · SHAP · Streamlit
>
> **Green-Material Inventory Blend Optimization** 2024 – 2025
>
> LOTTE Fine Chemical
>
> – Delivered one of three PoCs selected under the company-wide AI master plan, alongside raw-material price forecasting and an LLM-based service platform.
> – Developed an optimization approach for choosing blend combinations from green-material inventory.
>
> Conversational Service for Power Plant Maintenance History Power generation company 2025
>
> AI Master Plan and LLM Service PoC LOTTE Fine Chemical 2024 – 2025

MakinaRocks, AI Project Team Apr 2023 – Apr 2024

Part Leader / Senior ML Engineer / PO

Developed anomaly detection, control optimization, and guidance models as well as RAG-based services, and supported LLMOps and MLOps environment setup. Planned new projects, improved the traceability and reproducibility of project deliverables, researched ML technology trends, and hired talent.

> ML and Control Optimization for a Sinter Plant Steel maker 2023 – 2024
>
> Q&A Chatbot for a Manufacturing Equipment Diagnosis System 2023
>
> ML Models for Petrochemical Equipment Lifetime Prediction and Condition Diagnosis Petrochemical company 2023

Samsung Electro-Mechanics, Equipment Technology Research Institute Mar 2021 – May 2022

Principal Researcher

Built a software platform for data-driven anomaly detection and predictive maintenance and for running deep learning models. Analyzed and visualized manufacturing equipment performance and logistics data, and taught data analysis to employees.

> Visual Inspection Automation Software and ML Platform 2021 – 2022

KEPCO (Korea Electric Power Corporation), Data Science Lab Jul 2010 – Jun 2020

Senior Data Scientist

Researched anomaly detection algorithms, time-series forecasting, and deep learning, including sequence and generative models. Built the AI and data science roadmap, identified and planned new research projects, and planned research infrastructure and analytics environments.

> Anomaly Detection from Transformer Oil DGA Data 2019 – 2020
>
> Object Detection for Distribution Equipment Diagnosis 2019 – 2020
>
> Power Usage Pattern Analysis of EV Charging Stations in Jeju 2018 – 2019
>
> Solar Power Forecasting for Distribution Lines 2016 – 2019
>
> Commissioning and Performance Evaluation System for Frequency Regulation ESS 2015 – 2018

Optimal Control for Power Plant Boiler Systems 2013 – 2014

POSCO E&C, Energy Business Division Jan 2008 – Jul 2010

Automation Design Engineer

Designed electrical and automation systems for power and energy plants, designed and built data acquisition infrastructure, developed control algorithms, and supported commissioning.

04 OTHER PROJECTS

Agentic RAG Search Service for National Legal Information 2026

Lifestyle Guide Service Application GIVita 2022

05 PATENTS

Registered 7 · Filed 8

2026 Method and system for automatically generating P&ID work reports and verifying them against the original drawings using a vision-language model Filed

2020 Main steam temperature control apparatus and method Registered

2020 Apparatus and method for diagnosing power equipment Filed

2019 Traffic analysis apparatus for network-based control systems and operating method thereof Registered

2018 Control system for a hybrid energy storage system Filed

2018 Power plant simulator Registered

2017 Superheater temperature control method Registered

2017 Frequency control system Filed

2015 Steam turbine dynamics simulator, gas turbine dynamics simulator, and turbine generator control apparatus and method using the same Filed

2015 Model-based predictive control apparatus and method Filed

2014 Operation control method for an oxy-fuel combustion boiler Registered

2014 Air supply apparatus and air supply method for combustion equipment Registered

2014 Integrated control apparatus for a supercritical Brayton cycle with a variable heat source Filed

2014 Interface apparatus for a process control system Registered

2013 Lubricant supply apparatus Filed

06 SOFTWARE

– Image data preprocessing program for equipment recognition (deep learning object detection) (Python 3.6) Pending

– Automated data preprocessing program for training anomaly detection machine learning models (Python 3.6) Pending

2018 ESS control algorithm simulation program (Matlab Simulink) Registered

2018 Simulator model for verifying gas turbine combined-cycle plant operation Registered

2015 Frequency analysis program for proportional-integral-derivative control loops (Matlab) Registered

2015 Automatic PID controller tuning parameter calculation program for power generation control systems Registered

2015 Process identification program for proportional-integral-derivative controller design (Matlab) Registered

2015 Dynamic matrix-based predictive controller and simulator (C, C++) Registered

2014 PID controller tuning parameter calculation program (Matlab GUI) Registered

2014 Matrix computation program for multiple-input multiple-output (MIMO) control (C, C++) Registered

07 PAPERS

2026 A Novel Diagnostic Method on the Industrial Boilers Using Machine Learning: t-SNE, Auto-Encoder and LGBM D. J. Jang, M. J. Jeon, K. M. Yu, and M. C. Lee · Energies, vol. 19, no. 4, art. 874 · doi.org/10.3390/en19040874

2018 Lessons Learned from Energy Storage System Demonstrations for Primary Frequency Control K. Yu, I. Choi, and J. Woo · KEPCO Journal on Electric Power and Energy, vol. 4, no. 2, pp. 107-114 · doi.org/10.18770/KEPCO.2018.04.02.107

2015 Development of a DMC Block for Use with an RCP System and its Application Y.-S. Lee and K.-M. Yu · Journal of Institute of Control, Robotics and Systems, vol. 21, no. 9, pp. 827-835 · doi.org/10.5302/J.ICROS.2015.15.0086

08 SELECTED REPORTS

2026	Physical AI humanoid product (GenP) definition: common core and domain profiles	GenON
2025	Core generative AI technologies and the ecosystem for building services	PwC Consulting
2023	A survey of Transformer-based long-term time-series forecasting (LTFS) methods	MakinaRocks
2022	Case study of a personalized diet management service using CGM data	GIVita
2020	Deep learning for anomaly detection: unsupervised approach	KEPCO Data Science Lab
2018	The need for frequency regulation ESS as renewable generation grows	Korean Institute of Electrical Engineers (KIEE), The Proceedings of the KIEE, vol. 67, no. 3

+13 more reports at sguys99.github.io

09 TEACHING, ADVISORY & BOOKS

2020 - 2026	Visiting Researcher / Technical Advisor	Department of Safety Engineering, Incheon National University
2026	Lectures on LLMs, agents, and Harness Engineering for undergraduate and master's researchers	Engineering Education Center, Incheon National University
2025	AI training curriculum for consultants (design and lectures)	PwC Consulting
2014 - 2023	Hands-on course on control systems and data analysis	Korea Measurement & Control Technology Education Center
2023	Big data analyst training program	SK planet Tacademy
2023	Software version control with Git and GitHub (parts 1 and 2)	Koscom
2023, 2024	Advisory Committee Member, Future Education	Incheon National University
2019 - 2020	Data science lectures for new ICT employees	KEPCO Human Resources Development Institute
2022	Learning Git & GitHub with Python Code	Youngjin.com · Book
2021	Git for Visual Studio Users	WikiDocs · Book